

Lab 20: Making Statistical Claims

*Instructor: Jonathan Pipping-Gamón**Authors: JP, RB*

20.1 Interpret Your Earlier Models

In this lab, you will interpret findings from past labs. You may reuse code, tables, figures, and saved outputs from your earlier submissions. Unless you want to, you do not have to refit large models from scratch.

20.1.1 Lab 14 — Fully Bayesian Models

Write a short note to **the Dallas Cowboys coaching staff** as they prepare for a game against **the Philadelphia Eagles**.

Use one result from your Bayesian team-strength analysis to explain the difference in strength between the two teams and what that should mean for game planning. Your note should make clear what quantity the model actually estimated, how certain or uncertain that comparison is, and what the model does not capture well enough to settle a coaching decision by itself.

20.1.2 Lab 15 — Regularization and Ridge

Write a short note to **the general manager of the New York Knicks** about whether the team should seriously consider trading for **Mikal Bridges**.

Use one result from your player-impact analysis to explain what your offensive/defensive RAPM result says about Bridges as a trade target. Your note should make clear what kind of impact the metric is measuring, what it leaves out, and how much decision weight this result deserves in a trade discussion.

20.1.3 Lab 19 — Kernel Methods

Write a short note to **Liverpool's training staff**.

Use one result from your expected-goals analysis to explain which shots the team should prioritize in training and match preparation. Include analysis of both closer and further shots. Be clear about what the model is actually saying about shot quality, what was validated, and what coaching conclusions would go too far.